FOR AN ETALE MORPHISM THE UPPER SHRIEK IS THE DERIVED PULLBACK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FWI (sources/stacks-duality/pinned-original-source.tex line 5044).

Allowed standard prerequisites (editorial).

- An etale morphism is flat, syntomic and local complete intersection. Morphisms, Lemmas labelled morphisms-lemma-etale-syntomic and morphisms-lemma-etale-flat, and More on Morphisms, Lemma labelled more-morphisms-lemma-flat-lci, cited at proof lines 5050-5054. Makes the available results applicable.
- Invertibility of the shriek of the structure sheaf and the perfect comparison. Same chapter, Lemmas labelled lemma-lci-shriek and lemma-perfect-comparison-shriek, cited at proof lines 5054-5056 and 5060-5062. Reduce the statement to f^!O_Y = O_X and give invertibility.
- The diagonal of an etale morphism is an open immersion and the shriek of an open immersion is the pullback. Morphisms, Lemmas labelled morphisms-lemma-diagonal-unramified-morphism and morphisms-lemma-etale-smooth-unramified, cited at proof lines 5060-5064, and Same chapter, Lemma labelled lemma-shriek-open-immersion, cited at proof lines 5064-5066. Gives Delta^! = Delta^*.
- Composition of shrieks and base change of the shriek along a flat morphism. Same chapter, Lemmas labelled lemma-upper-shriek-composition and lemma-base-change-shriek-flat, cited at proof lines 5066-5072. Give the composition identities used in the chain.

Source reference. Original reference 0F42: sources/stacks-duality/pinned-original-source.tex, source lines 3953–3953. Same-chapter pointer into the author environment situation at source line 3953; invoked at proof line None of the counted proof.

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It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02GQ: sources/stacks-chow/tags.txt, source lines 3237–3237. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 5053 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02GS: sources/stacks-chow/tags.txt, source lines 3239–3239. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 5054 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 069K: sources/stacks-chow/tags.txt, source lines 8044–8044. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 5055 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B6U: sources/stacks-duality/pinned-original-source.tex, source lines 4798–4798. Same-chapter pointer into the author environment lemma at source line 4798; invoked at proof line 5056 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B6V: sources/stacks-duality/pinned-original-source.tex, source lines 4909–4909. Same-chapter pointer into the author environment lemma at source line 4909; invoked at proof line 5058 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02GE: sources/stacks-chow/tags.txt, source lines 3225–3225. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 5068 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02GK: sources/stacks-chow/tags.txt, source lines 3231–3231. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 5069 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AU0: sources/stacks-duality/pinned-original-source.tex, source lines 4568–4568. Same-chapter pointer into the author environment lemma at source line 4568; invoked at proof line 5070 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ATX: sources/stacks-duality/pinned-original-source.tex, source lines 4161–4161. Same-chapter pointer into the author environment lemma at source line 4161; invoked at proof line 5071 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0E9U: sources/stacks-duality/pinned-original-source.tex, source lines 4997–4997. Same-chapter pointer into the author environment lemma at source line 4997; invoked at proof line 5073 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0F42 let $f : X \to Y$ be an étale morphism of FTS_S. Then $f^! \cong f^*$ as functors on $D^+_{QCoh}(\mathcal{O}_Y)$.*

Proof. We are going to use that an étale morphism is flat, syntomic, and a local complete intersection morphism (Morphisms, Lemma 02GQ and 02GS and More on Morphisms, Lemma 069K). By Lemma 0B6U it suffices to show $f^!\mathcal{O}_Y = \mathcal{O}_X$. By Lemma 0B6V we know that $f^!\mathcal{O}_Y$ is an invertible module. Consider the commutative diagram

$$\begin{array}{ccc} X \times_Y X & \xrightarrow{p_2} & X \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle f} \\ X & \xrightarrow{f} & Y \end{array}$$

and the diagonal $\Delta : X \to X \times_Y X$. Since Δ is an open immersion (by Morphisms, Lemmas 02GE and 02GK), by Lemma 0AU0 we have $\Delta^! = \Delta^*$. By Lemma 0ATX we have $\Delta^! \circ p_1^! \circ f^! = f^!$. By Lemma 0E9U applied to the diagram we have $p_1^!\mathcal{O}_X = p_2^* f^!\mathcal{O}_Y$. Hence we conclude

$$f^!\mathcal{O}_Y = \Delta^! p_1^! f^!\mathcal{O}_Y = \Delta^*(p_1^* f^!\mathcal{O}_Y \otimes p_1^!\mathcal{O}_X) = \Delta^*(p_2^* f^!\mathcal{O}_Y \otimes p_1^* f^!\mathcal{O}_Y) = (f^!\mathcal{O}_Y)^{\otimes 2}$$

where in the second step we have used Lemma 0B6U once more. Thus $f^!\mathcal{O}_Y = \mathcal{O}_X$ as desired. □